

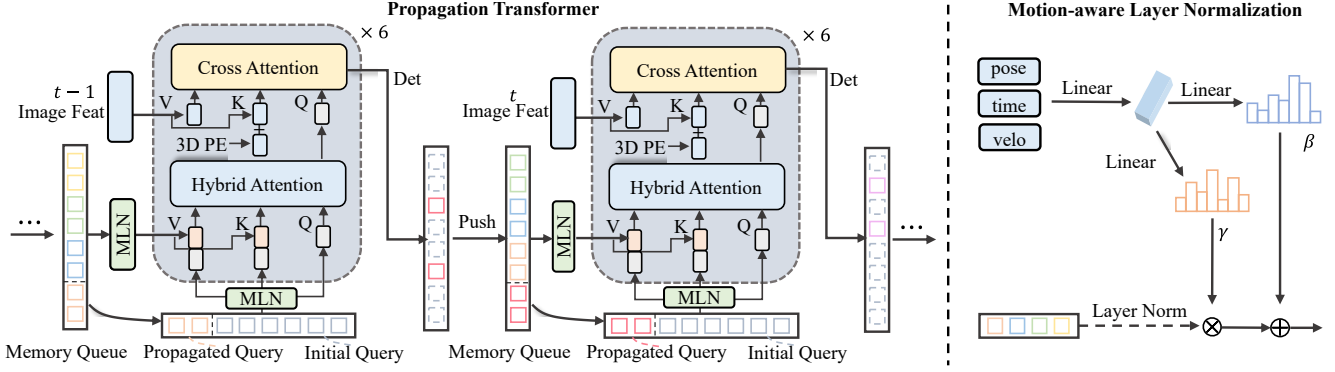


Figure 4. The details of the propagation transformer and motion-aware layer normalization. In the propagation Transformer [43], object queries interact with hybrid queries and image features iteratively. The motion-aware layer normalization encodes the motion attributes (ego pose, timestamps, velocity) and performs a compensation implicitly. Rectangles of varying hues symbolize queries from distinct frames, gray rectangles represent initialized queries of current frame, dashed rectangles correspond to background queries.

$$\tilde{F}_{obj}^t = \varphi(\tilde{F}_{obj}^{t-1}, F_{obj}^t) \quad (7)$$

4. Method

4.1. Overall Architecture

As illustrated in Fig. 3, StreamPETR is built upon end-to-end sparse query-based 3D object detectors [30, 47]. It consists of an image encoder, a recursively updated memory queue, and a propagation transformer [43]. The image encoder is a standard 2D backbone, which is applied to extract semantic features from multi-view images. Then the extracted features, information in the memory queue, and object queries are fed into the propagation transformer to perform the spatial-temporal interaction. The main difference between StreamPETR and single-frame baseline is the memory queue, which recursively updates the temporal information of object queries. Combined with the propagation transformer, the memory queue can propagate temporal priors from previous to current frames efficiently.

4.2. Memory Queue

We design a memory queue of $N \times K$ for effective temporal modeling. N is the number of stored frames and K is the number of objects stored per frame. According to the experience, we set $N = 4$ and $K = 256$ (ensuring high recall in complex scenarios). After the preset time interval τ , the relative time interval Δt , context embedding Q_c , object center Q_p , velocity v , and ego-pose matrix E of selected object queries are stored in memory queue. Specifically, the above information, corresponding to foreground objects (with top- K highest classification score), is selected and pushed into the memory queue. The entrance and exit of the memory queue follow the first-in, first-out (FIFO) rule. When information from a new frame is added to the memory queue, the oldest is discarded. Actually, the proposed memory queue is highly flexible and customized, users can

freely control the maximal memory size $N \times K$ and saving interval τ during both training and inference.

4.3. Propagation Transformer

As illustrated in Fig. 4, the propagation transformer consists of three main components: (1) the motion-aware layer normalization module implicitly updates the object state according to the context embedding and motion information recorded in the memory queue; (2) the hybrid attention replaces the default self-attention operation. It plays the role of temporal modeling and removing duplicated predictions; (3) the cross-attention is adopted for feature aggregation. It can be replaced with an arbitrary spatial operation to build the relationship between image tokens and 3D object queries, such as global attention in PETR [30] or sparse projective attention in DETR3D [47].

Motion-aware Layer Normalization is designed to model the movement of objects. For simplicity, we take the transformation process from the last frame $t-1$ as the example and adopt the same operation for other previous frames. Given the ego pose matrix from the last frame E_{t-1} and current frame E_t , the ego transformation E_{t-1}^t can be calculated as:

$$E_{t-1}^t = E_t^{inv} \cdot E_{t-1} \quad (8)$$

Assume that objects are static, 3D centers Q_p^{t-1} in memory queue can be explicitly aligned to the current frame, which is formulated as:

$$\tilde{Q}_p^t = E_{t-1}^t \cdot Q_p^{t-1} \quad (9)$$

where $\tilde{Q}_p^t$ is the aligned centers. Motivated by the task-specific control in generative model [9, 40, 55], we adopt a conditional layer normalization to model the movement of the objects. As shown in Fig. 4, the default affine transformation in layer normalization (LN) is closed. The motion attributes $(E_{t-1}^t, v, \Delta t)$ are flattened and converted to affine